

Parker Barton

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Mechanical engineering junior targeting R&D — CAD, FEA, CAM, machining, PCB layout, firmware, field testing.

EDUCATION

Purdue University

West Lafayette, IN

B.S. Mechanical Engineering, Minor in Management

Aug 2024 – May 2028

- Coursework: Mechanics of Materials, Fluid Mechanics, Machine Design, System Modeling and Control, Materials Science, Thermodynamics, Dynamics, Statics, Circuits, Manufacturing, Differential Equations, Programming (C, MATLAB)

EXPERIENCE

Robotics Engineering Intern

May 2026 – Aug 2026

Rock Rose Risk

Remote

- Sole mechanical and electrical designer for RoseBud, an autonomous rocker-bogie rover that scans residential property into 3D models for wildfire risk assessment; delivered three design generations and multiple field units
- Authored the sensing feasibility study that set program direction, scoring six LiDAR and photogrammetry systems across six criteria; eliminated a cheaper quadruped on data sovereignty and a drone on FAA Part 107 operating cost
- Designed a 2 oz copper PCB in KiCad (RP2040, dual H-bridge drivers, UBEC rails, 4S BMS) that consolidated an 80+ wire spliced harness onto one board and cut electronics footprint by 2 in
- Engineered a USB-C charge chain and specified an 88.8 Wh pack to stay under the 100 Wh air-transport limit
- Converted printed structural members to 6061 plate and sealed ball bearings after field testing, eliminating rocker-arm torsion and steering backlash across six wheel pods
- Field-tested over sand, rubble, and 45° grades; diagnosed transit fracture of 20 mm PETG joints and designed/manufactured 6061 replacements with rotomolded box and custom foam packaging.

Data Management Intern

May 2025 – Aug 2025

World Insurance Associates

Iselin, NJ

- Standardized client records across Excel, EPIC, and BenefitPoint with sorting rules that improved data accuracy and reduced account retrieval time

IT Support

May 2024 – Aug 2024

Brentwood Academy

Franklin, TN

- Maintained and upgraded campus IT infrastructure, including physical server and network switch replacement, alongside daily support for staff and students.

LEADERSHIP

Vice President, ACM / Mechanics Lead, Event Coordinator, SigBots

Aug 2024 – Present

Association for Computing Machinery, Purdue University

West Lafayette, IN

- Own organizational structure, a \$40,000 budget, and manufacturing for a 60+ member interdisciplinary robotics team; secure the industry sponsorships that fund it
- Lead mechanical design for the 24" VEX U robot in Inventor; designed 20+ competition robots using custom aluminum plate chassis and parametric mechanisms
- Team awarded the **2026 VEX U World Excellence Award**, the top overall award in VEX Robotics; program previously ranked #2 globally
- Built CAD, manufacturing, and documentation workshops onboarding 30+ new members annually, and directed a sponsor-funded renovation of a 15-year-old lab space

Structures & Manufacturing, IREC Team

Aug 2026 – Present

Purdue Space Program

West Lafayette, IN

- Structures and manufacturing for the 10,000 ft class competition vehicle, modeling airframe and structural components in Onshape
- Machine flight hardware on HAAS VF-4 and UMC-500 mills at the Bechtel Innovation Design Center
- Model vehicle trajectory and motor performance in OpenRocket and openMotor to support structural sizing

Team Captain, VEX Robotics Team 9364 (Iron Eagles)

2020 – 2024

Four seasons leading mechanical design, programming, CAD, and documentation

Franklin, TN

- Earned 60+ competition awards, 8 Signature Event championships, and two World Championship division finals

PROJECTS

- Bipedal Balancing Robot** | *Fusion 360, Raspberry Pi 5, RP2040, C++* 2026
- Designed every joint around a 1.2in structural ring that carries bending and radial load into the shell rather than the servo spline, and routes downstream wiring through the axis of rotation
 - Added hard mechanical stops to the leg geometry, dropping standing idle current from ~5 A to ~0.1 A and removing the need for additional voltage regulation
 - Split control between an RP2040 running a hard 400 Hz balance loop and a Pi 5 handling perception, with a 500 ms watchdog that keeps the robot upright through a host reboot
 - Selected a 12 g digital gripper servo over a 55 g analog unit, removing 0.86 kg-cm from the shoulder; spring-closed design holds payload at zero current
- PostIt - Voice Assistant** | *Raspberry Pi 5, Python, I2S audio* 2026
- Built an end-to-end voice pipeline (custom wake word, local speech recognition, on-device synthesis) with calendar, mail, music, and camera integration
 - Cut speech synthesis latency from ~10 s to ~0.2 s by loading the engine in-process, and reduced cached conversational turns to ~0.6 s
 - Designed a two-revision printed enclosure with a 2.8" touch panel and nine-screen UI; runs headless and recovers on power loss
- MySwing - Smart Golf Training Grip** | *ESP32-C3, IMU, BLE* (Provisional Patent Filed) 2025 – 2026
- Filed a provisional patent on a portable weighted training grip with embedded motion tracking and haptic tempo feedback
 - Developed firmware streaming IMU data over BLE to a web interface estimating club speed, face angle, and shot shape; segmented swings by phase against 30+ logged sessions
 - Machined and epoxy-bonded an aluminum shaft and 1018 steel end mass on a lathe after selecting an IMU that stayed coherent at full swing speed
- CNC Milled Golf Clubs - Putter & 7 Iron** | *NX, Fusion 360, HAAS UMC-500* 2025 – 2026
- Designed parametric heads to USGA equipment rules: five 6061 putters held to a 340–360 g target, and a 1018 steel iron tunable to a player's miss
 - Raised moment of inertia from 2,700 to 3,294 g-cm² (+22%) through heel mass placement; FEA showed 0.12–0.18 mm peak face deflection under impact
 - Programmed 5-axis toolpaths and corrected multi-axis translation accelerations, cutting cycle time from ~5 h to ~2 h per head (-60%)
 - Designed the shaft interface perpendicular to the 4° lofted face as a repeatable second-operation datum, eliminating soft jaws; finished with a self-built Type II anodizing line
- Cast Bronze Wristwatch** | *SLA wax printing, investment casting* 2025
- Designed a three-part case around a purchased movement, tolerancing for casting shrinkage, post-machining, and serviceability
- 3D-Printed Dorm Hardware - Design Through Sales** | *Fusion 360, FDM* 2024 – 2025
- Designed snap-fit fixtures for dorm furniture, tuning clearance and print orientation to stop cracking; batch-produced and sold across campus

TECHNICAL SKILLS

Machining & Fabrication: 1–5 axis CNC (HAAS UMC-500), manual mill/lathe, waterjet, laser, FDM/SLA, investment casting, anodizing

CAD & Analysis: NX, SolidWorks, Fusion 360, Inventor, Ansys/SpaceClaim; parametric modeling, GD&T, FEA, mass properties

Electronics & Software: KiCad schematic capture and PCB layout, power and rail design, BMS and LiPo charging, soldering and rework; Python, C/C++, MATLAB, JavaScript, embedded control loops

Engineering Process: Design for manufacturing, decision matrices, BOM and cost modeling, design reviews, technical documentation